

BANK CHURN · A BUSINESS PERSPECTIVE

Why Are Our Customers Leaving?

A churn & retention analysis of 10,000 retail-banking customers

20.4% churn · **€186M deposits at risk** · **2,037 customers lost**

Team & Mentor

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Dataset

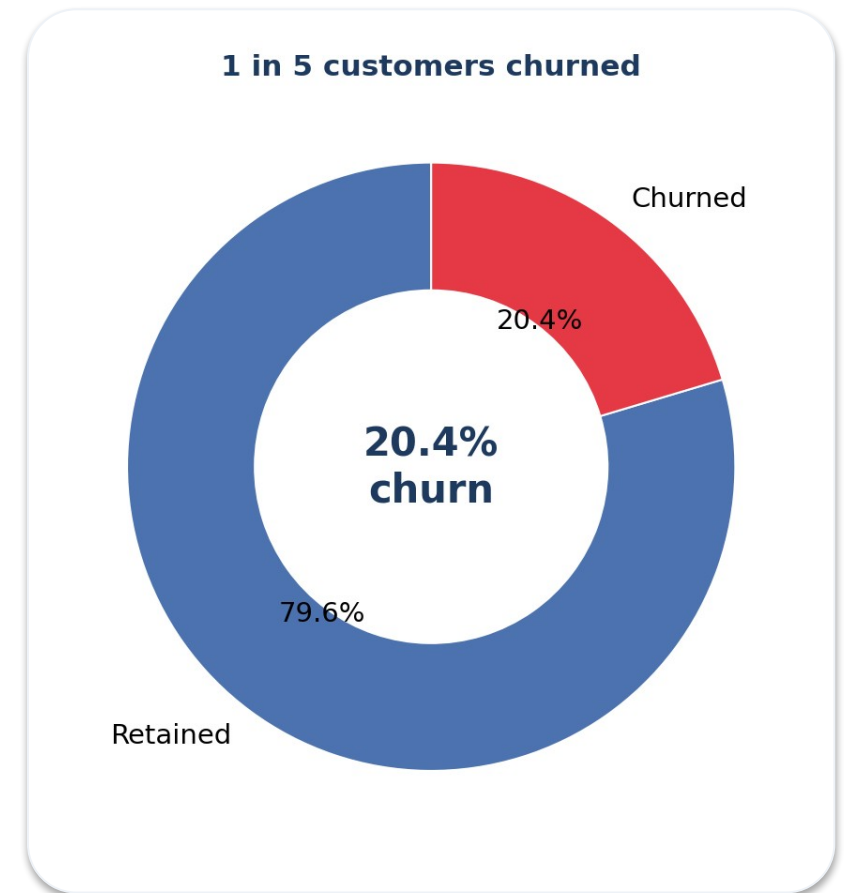
Bank Churn — 10,000 retail-banking customers

Tools

Python (pandas) · cleaning & analysis | Plotly · interactive dashboard | PowerPoint

The Domain — Retail Banking Churn

- ▶ **Churn** = a customer leaves the bank (closes their relationship). Our flag: Exited = 1.
- ▶ **Why it matters:** acquiring a new customer costs far more than keeping one — and a churned customer takes their deposits and fee income with them.
- ▶ **The base:** 10,000 customers across France, Germany and Spain, with demographics, account balances, products held and activity.
- ▶ **Our questions:** How many leave? WHO leaves, and WHY? And WHERE should the bank intervene first?



KPIs & Success Metrics



Thought process

- ▶ **Churn rate is the north-star metric** — the single number that tells us whether we are winning or losing customers.
- ▶ **Balance at risk** translates churn into euros, so the business feels the impact (€186M $\approx$ a quarter of all deposits).
- ▶ **Active-member rate & products-per-customer** are the levers we can actually pull — engagement and product fit drive retention.
- ▶ **Success** = move churn from 20% $\rightarrow \leq 15\%$ and lift retention above 85% over the next year.

Data Preparation & Assumptions

- ▶ **Source:** one Excel workbook, two sheets (Customer_Info + Account_Info), joined on CustomerId.
- ▶ **Standardised geography** — France was split across 'France/French/FRA' → unified.
- ▶ **Fixed types** — stripped '€' from Balance & EstimatedSalary (text→number); Yes/No → 1/0.
- ▶ **Removed duplicates**, imputed 3 missing ages (median), reconciled a duplicated Tenure column.
- ▶ **Filters/assumptions:** snapshot data (no time dimension); full customer base analysed, no sampling.

Data-quality catch

'HasCrCard' was identical to 'IsActiveMember' for every row — a corrupted column. Detected and restored to authentic values.

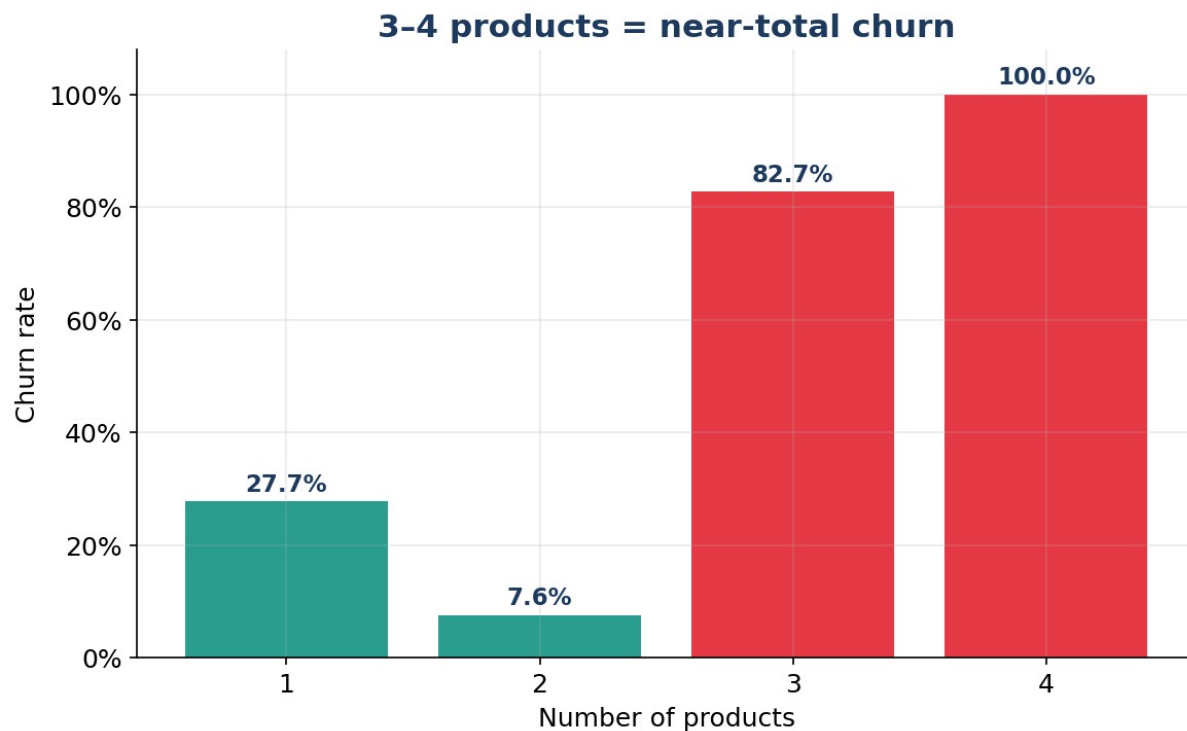
Validation vs. published reference

Rows	10,000 ✓
Missing values	0 ✓
Churn rate	20.37% ✓
Geography mix	match ✓
Total balance	€0.77bn ✓

The Dashboard — One Page, Six Lenses



Insight 1 — The Product Paradox

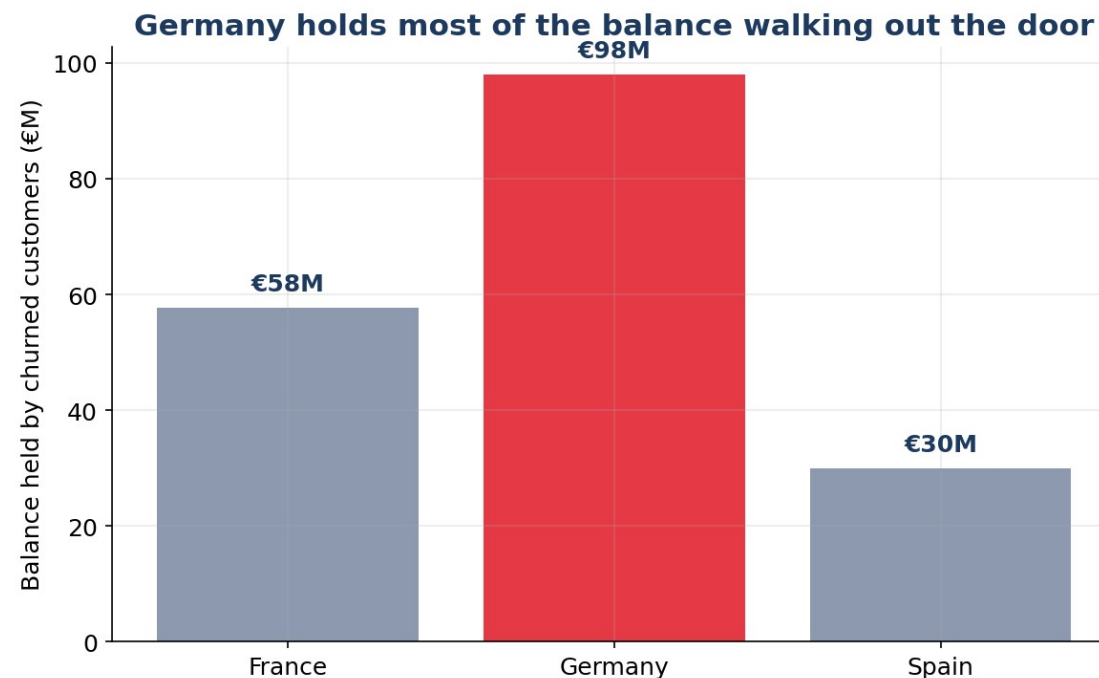
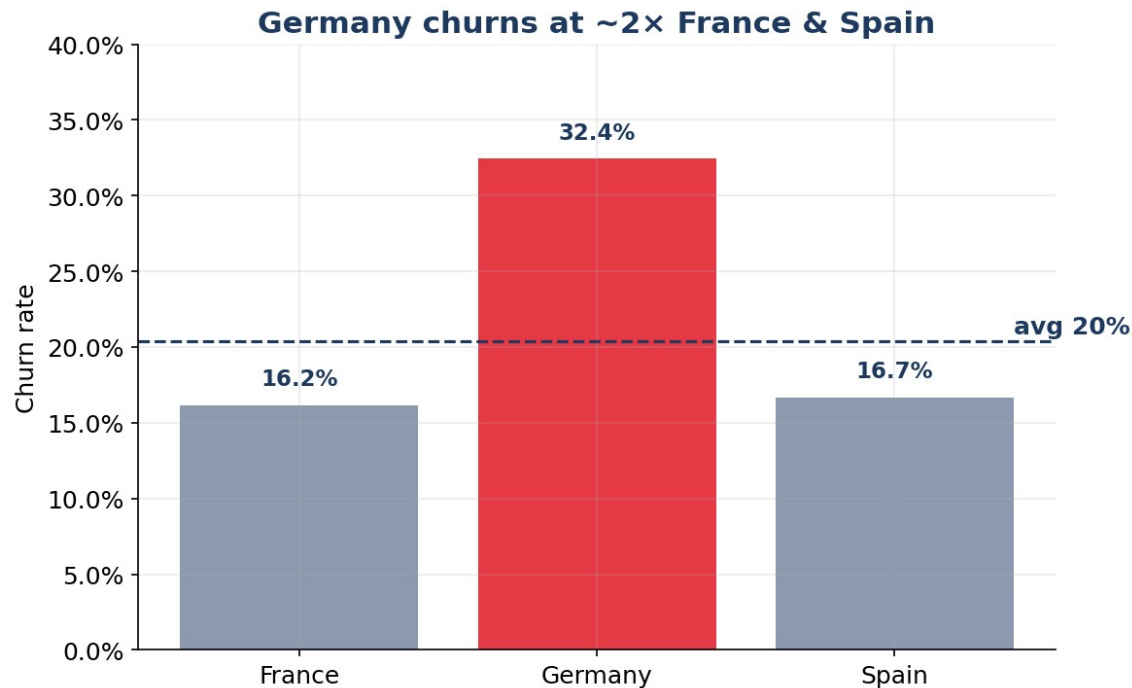


- ▶ **2 products is the sweet spot** — just 7.6% churn. These customers are engaged and sticky.
- ▶ **3-4 products = near-certain exit** — 83% and 100% churn. The 326 customers here are almost all gone.
- ▶ **Half the base (51%) holds a single product** and churns at 27.7%.
- ▶ **Read:** piling products onto unhappy customers backfires; the goal is the right second product, not more products.

1 → 2 products

moving single-product customers to two is the biggest retention prize.

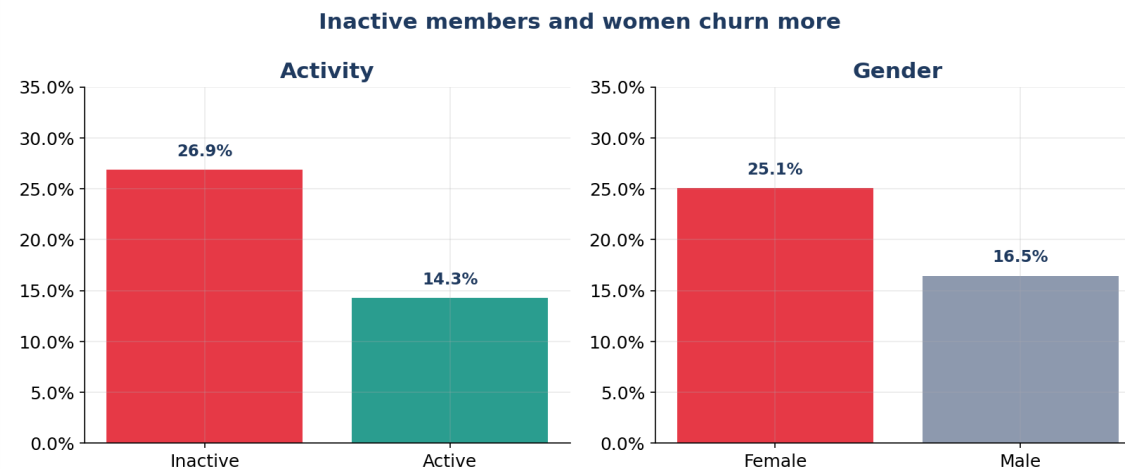
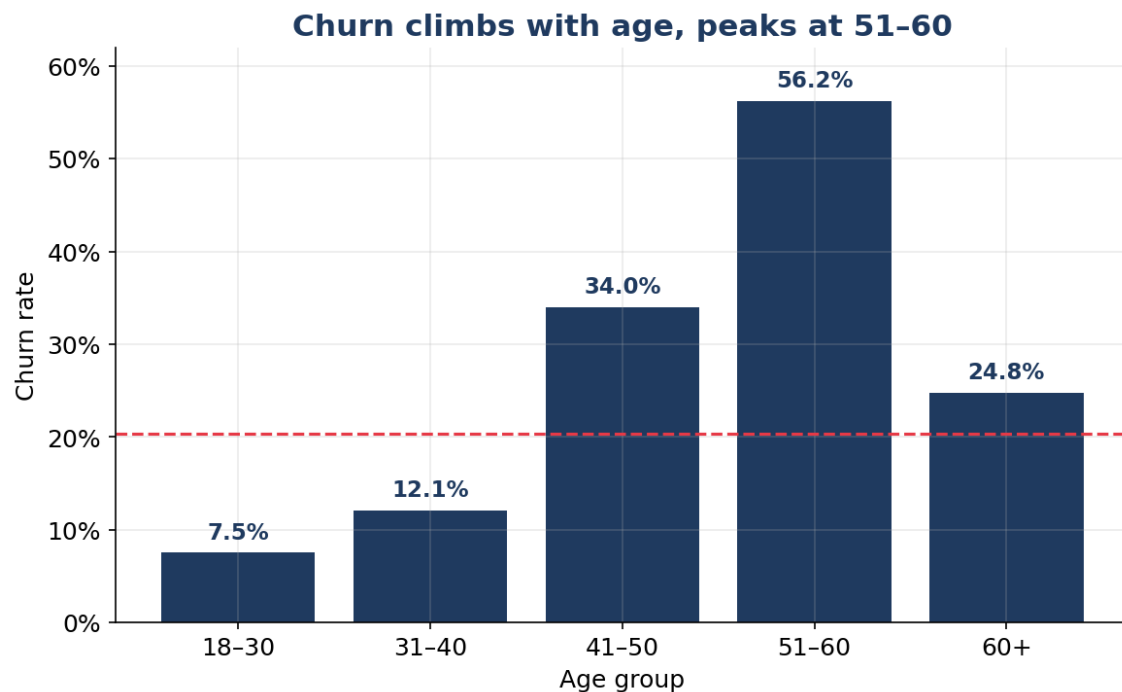
Insight 2 — Germany & the Deposit Drain



Germany: 32% churn • €120k avg balance • €98M walking out the door

Germany churns at twice the rate of France & Spain, yet holds ~2× the average balance — so it is both our biggest leak and our most valuable one. Priority market #1.

Insight 3 — Age & Engagement



2,521 customers are BOTH inactive and single-product — and 37% of them churn

Age 51-60 churns at 56%; inactive members at 27% (vs 14% active); women at 25% (vs 17%). The inactive + single-product overlap is the single most actionable at-risk segment.

Equally Useful — What Does NOT Drive Churn

Three things the bank might expect to matter — but the data says don't:

Tenure

≈ 19–21% churn at every tenure

Loyalty length is flat — long-tenured customers leave just as often.

Credit score

22% (poor) vs 19% (good)

Only a faint signal — not a useful targeting variable.

Has credit card

20.8% vs 20.2%

No effect. (This is also where we caught the corrupted column.)

Focus the retention budget where it counts

products, geography, age and engagement — not tenure, credit score or card ownership.

Recommendations

1

Launch a Germany retention task force

32% churn + high balances → review pricing, product fit and local service. Biggest single opportunity.

2

Fix the multi-product problem

Audit why 3-4-product customers leave (fees? mis-selling?) before any further cross-sell to them.

3

Move single-product customers to two

Targeted, value-led second product — 2 products is the 7.6%-churn sweet spot. Largest reachable group.

4

Re-activate the inactive

Engagement campaign for the 2,521 inactive single-product customers churning at 37%.

5

Protect the 45-60 segment

Proactive relationship outreach for older, higher-churn, higher-balance customers (and women).

CONCLUSION

1 in 5 customers leave — but the churn is concentrated and addressable.

- ▶ Churn is driven by product fit, geography, age and engagement — not by tenure or credit history.
- ▶ Germany, multi-product and inactive single-product customers are the clear priority segments.
- ▶ Cutting churn from 20.4% to 15% would retain ~537 more customers each cycle and protect tens of €M in deposits.

Target: churn $\leq$ 15% • retention $\geq$ 85%

Questions?

Thank you · Bank Churn — A Business Perspective

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